

Bùi Nguyễn Thắng

Fullstack Developer · .NET · Product · Middle+

Building products with C#/.NET — strongest on backend, distributed systems and digital signing, and comfortable on the front end too; owning each product from analysis through to live users.

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PROFESSIONAL SUMMARY

Fullstack developer with 5+ years of experience, focused on backend and distributed systems in C#/.NET while also building front ends with ReactJS when a product needs it. Solid hands-on experience with RESTful APIs, microservices, asynchronous processing, caching, messaging, NoSQL/SQL databases, authentication, digital signatures, and containerized environments. Regular day-to-day work with RabbitMQ, Redis, Cassandra/ScyllaDB, Docker, HAProxy, Nginx, OpenTelemetry, and Jenkins.

A product developer rather than a short-term project one: I stay with the same product across many releases, from business analysis and design through development, operations, and supporting real users. Experienced in designing high-performance, scalable systems and troubleshooting production issues across application, infrastructure, and networking layers. Currently going deeper into system design, architecture, scalability, and technical decision-making.

WORK EXPERIENCE

Fullstack Developer · BNNSoft

2020 — Present

- Built and maintained .NET backend services (ASP.NET Core, Minimal API, ABP Framework) serving REST APIs and realtime features over SignalR.
- Built admin web panels for each system — BHXH, e-invoice, e-contract, HRM — for internal operations: service configuration, user permissions, record lookup, processing-status monitoring, and customer support.
- Designed asynchronous processing pipelines with RabbitMQ and background workers; improved throughput with Redis caching and rate limiting.
- Modelled data and optimised queries on Cassandra/ScyllaDB and MySQL/MariaDB for write-heavy workloads.
- Implemented PDF digital signing (PAdES) using PKI, X.509, HSM, PKCS#7 and TSA.
- Ran containerised infrastructure: Docker Compose, HAProxy/Nginx/Traefik, and Jenkins CI/CD on Linux.
- Integrated hardware devices: thermal printers (ESC/POS) and attendance machines, syncing data both ways between devices and the system.
- Set up observability with OpenTelemetry plus Jaeger/Grafana/Loki/Tempo/Prometheus to trace and debug production issues.

KEY PROJECTS

BHXH Social Insurance Integration System

bhdv.vin-bhxx.com

- Developed services for digitally signing XML documents and submitting electronic procedures to the Vietnam Social Security system.
- Implemented XML signing and certificate-based authentication.
- Integrated APIs for submitting and processing BHXH procedures.
- Designed asynchronous processing using RabbitMQ for high-volume requests.
- Applied Redis for caching and token management.
- Worked with distributed workers and load balancing to improve scalability and reliability.

XML Signing · X.509 · RabbitMQ · Redis · Distributed Workers · Load Balancing

CRM Enterprise Service Management Platform

crm.vin-ra.vn

- Developed a CRM platform managing multiple enterprise services, including BHXH, HSM, and HĐĐT.
- Designed service management and integration architecture for different external providers.
- Implemented authentication, authorization, token management, and API integrations.
- Applied Redis, RabbitMQ, and distributed processing to improve system performance and scalability.

ABP Framework · ASP.NET Core · JWT · Redis · RabbitMQ · REST API

HĐĐT Electronic Invoice System

hddt.vin-hoadon.com

- Developed electronic invoice services integrated with the Vietnam Tax Authority.
- Implemented XML digital signing and PDF digital signing.
- Worked with PKI, X.509 certificates, RSA, SHA-256, PKCS#7, HSM, PAdES, and TSA.
- Developed APIs for submitting electronic invoices and receiving processing results from tax systems.
- Optimized asynchronous processing and high-concurrency workloads.

ASP.NET Core · C# · RabbitMQ · Redis · REST API · PKI / X.509 · PAdES / TSA

HRM Human Resource Management System

hrm-bnn.hopdong.id.vn

- Developed HR management functionality and integrated with employee attendance machines.
- Implemented synchronization of attendance data between physical devices and the HRM system.
- Developed APIs and background services for data synchronization and processing.
- Worked with scheduling, asynchronous processing, and database optimization.
- Integrated thermal printers (ESC/POS) to print slips and forms directly from the system.

ASP.NET Core · Background Workers · Scheduling · MySQL · REST API · ESC/POS · TCP/IP

HĐĐT-LĐ Electronic Contract Platform

admin.hopdong.id.vn

- Built an electronic contract platform for enterprises and employees to sign online: create contracts from templates, invite signers, and track each party's status.
- Integrated digital signing via HSM (server-side signing for the enterprise) and USB token (client-side signing through a PKCS#11 plugin).
- Integrated VNeID for employees: identity verification and contract signing with a personal digital signature, with no dedicated signing device needed.
- Signs XML or PDF depending on the contract template format — PAdES for PDF, XMLDSig for XML, with TSA timestamping.
- Designed ordered multi-party signing flows, placing signatures at the right position and verifying document integrity and certificate validity after each turn.
- Recorded a full audit trail of the signing process and exported evidence for later verification.

ASP.NET Core · HSM / PKCS#11 · USB Token · VNeID API · PAdES / XMLDSig · TSA · PKI / X.509

VNeID Digital Signature Integration with VNeID

- Developed digital-signature functionality integrated with VNeID.
- Implemented certificate-based signing and document processing workflows.
- Worked with digital signature APIs and secure authentication flows.
- Integrated signing services into enterprise applications.

VNeID API · X.509 · Digital Signature · Secure Auth

TECHNICAL SKILLS

C# / .NET — core focus

C# · .NET 8/9 · ASP.NET Core · Minimal API · ABP Framework · SignalR · async/await & Task · Background workers & hosted services · Dependency injection · Middleware & filters

Backend & Distributed Systems

REST API · JWT · RabbitMQ · Redis · Concurrency · Rate limiting · Distributed processing & load balancing

Performance & Architecture

API performance optimization · Caching strategies · Database modeling & query optimization · Distributed system design · Concurrency & scalability · High availability and load balancing

Databases

Cassandra · ScyllaDB · MySQL · MariaDB · SqlSugar ORM

DevOps & Infrastructure

Docker · Docker Compose · Jenkins · Git · HAProxy · Nginx · Traefik · Linux/Ubuntu · Certbot

Observability

OpenTelemetry · Jaeger · Grafana · Loki · Tempo · Prometheus

Hardware Integration

Thermal printers (ESC/POS) · Attendance machines · Device data synchronisation · TCP/IP & serial communication

Frontend

JavaScript · TypeScript · ReactJS · React Native · Bootstrap · Tailwind CSS

Digital Signature (niche area)

PKI / X.509 · RSA / SHA-256 · HSM / PKCS#7 · PDF & XML signing (PAdES, TSA) · OpenSSL

AI-ASSISTED DEVELOPMENT

- Actively use AI-assisted coding to accelerate development, debugging, refactoring, code review, and technical research.
- Use AI Skills and MCP (Model Context Protocol) to integrate AI with development tools and engineering workflows.
- Build AI-assisted workflows for codebase analysis, documentation, troubleshooting, and repetitive development tasks.
- Leverage AI to improve productivity while maintaining control over architecture, implementation, testing, and code quality.

PERSONALITY & INTERESTS

Personality

Easy-going, easy to work with · Takes initiative · Careful with details · Keen on learning new tech

Interests

Gaming · Movies · Exploring new tech